

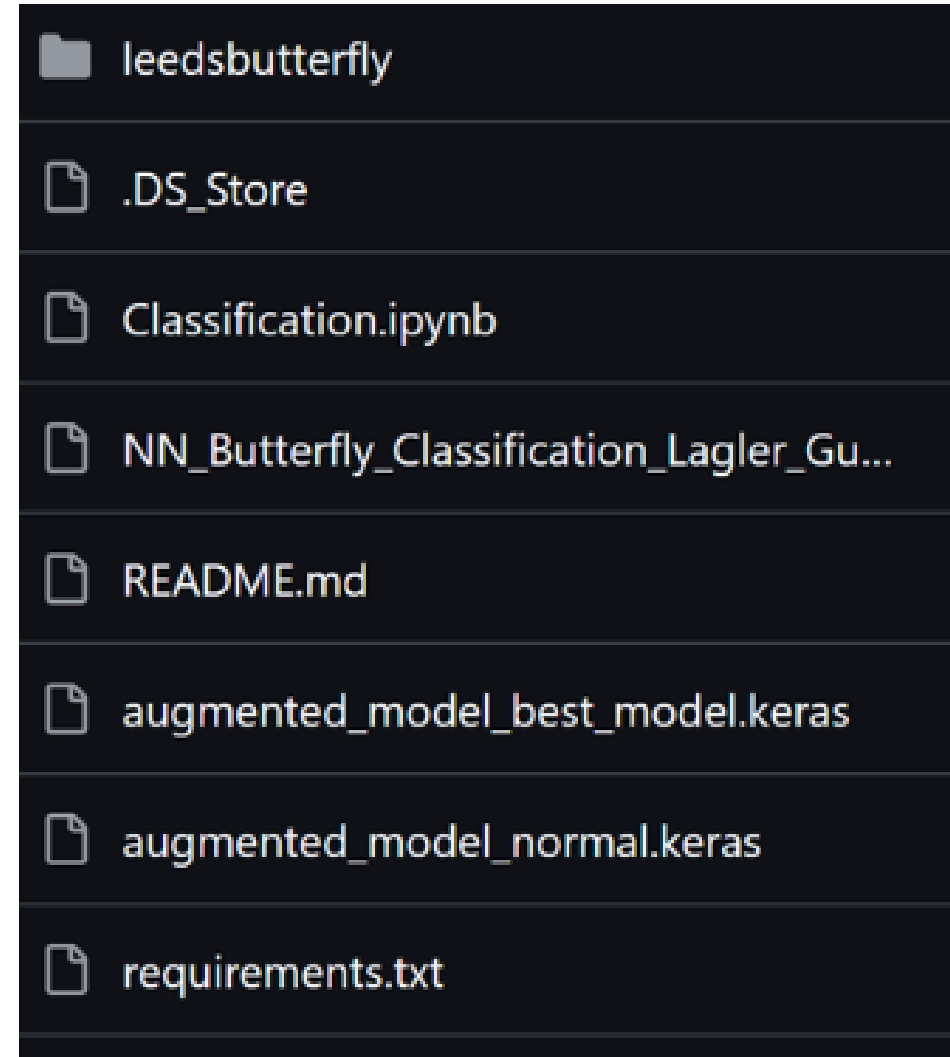
A 3D rendering of a warehouse conveyor belt system. Several cardboard boxes are positioned on the belt, which is flanked by blue guides. Red laser lines are projected across the scene, creating a grid pattern on the floor and highlighting the boxes. The perspective is from a low angle, looking down the length of the conveyor.

Butterfly Image Classifier

Kai Aebli, Mariia Gurova & Pia Lagler
Developing Software as a Product: V5_15 SS26

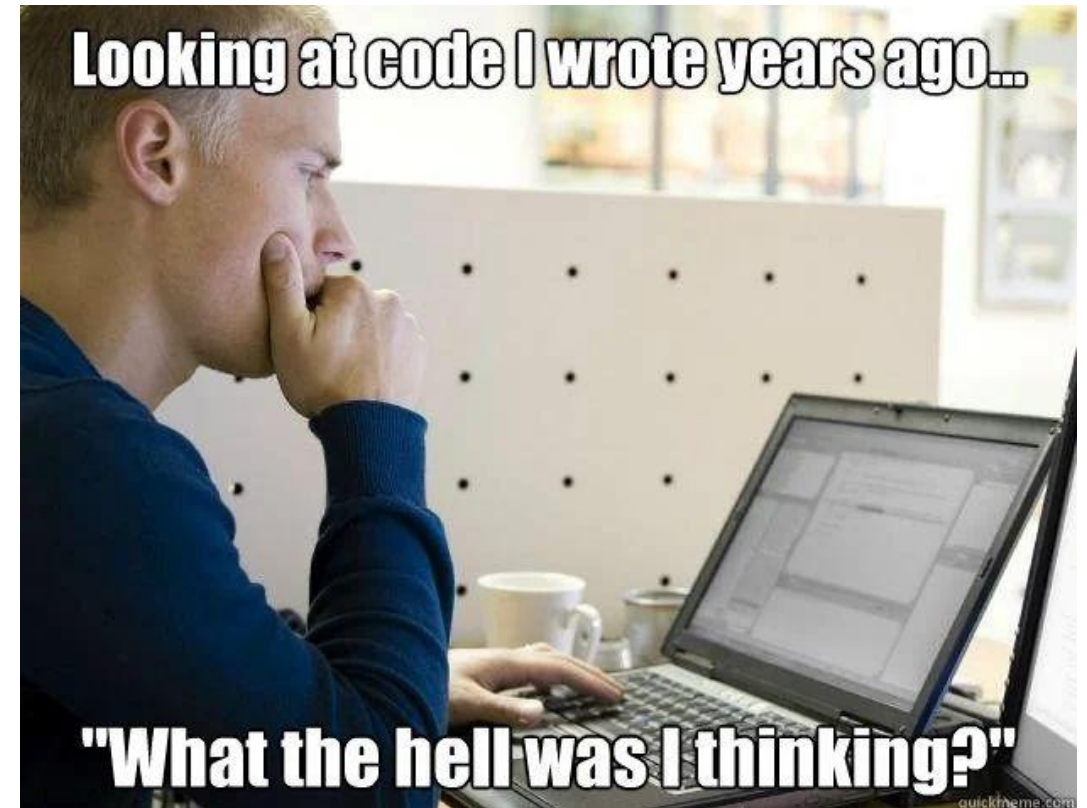
Initial Project State

- Monolithic structure – all files in one directory
- Main logic in .ipynb script
- Missing tests, CI/CD & license
- Dataset stored in repository
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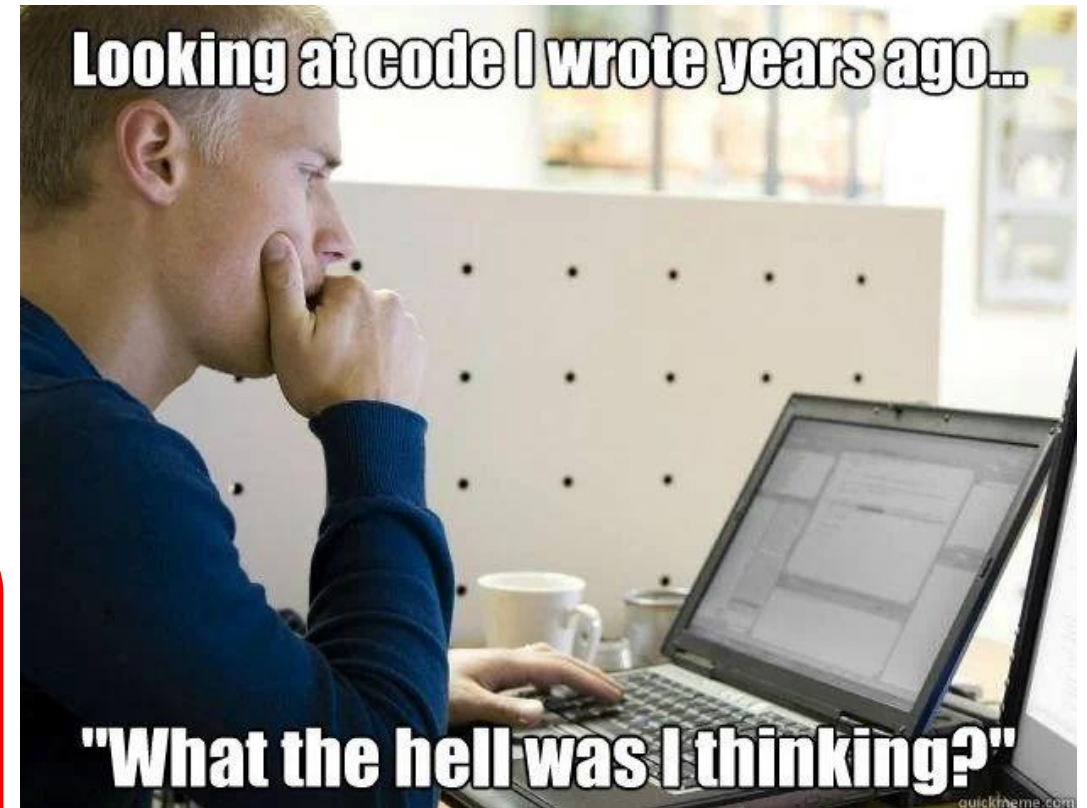


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Goal:

- Transform prototype into a software product
- Improve structure & maintainability
- Implement CI/CD, testing & coverage, workflows, branch collaboration



Refactoring & Setup

Cleaned code + meaningful comments

Automatic data download first

Organized project structure (scripts/, tests/, results/, etc)

Improved readability & naming

Reduced code duplication

Modular scripts in classes

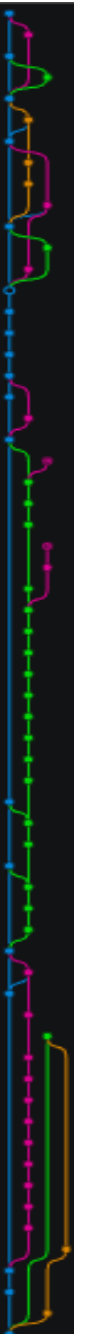
.toml for environment

Added license with MIT standard

```
✓ BUTTERFLY-IMAGE-CLASSIFICATION
> .github
> .vscode
> models
> results
> scripts
> tests
❖ .gitignore
! codecov.yml
📄 LICENSE
🔗 main.py
⚙️ pyproject.toml
📖 README.md
```

Git & Collaboration Workflow

- Main branch protection enabled
- Development performed in the isolated branches
- All changes submitted through PR & issues linked to them
- Required PRs before merging
- Revision from a team member before merge
- Project created for tracking TODOs

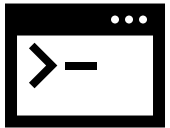


Profiling & Optimization

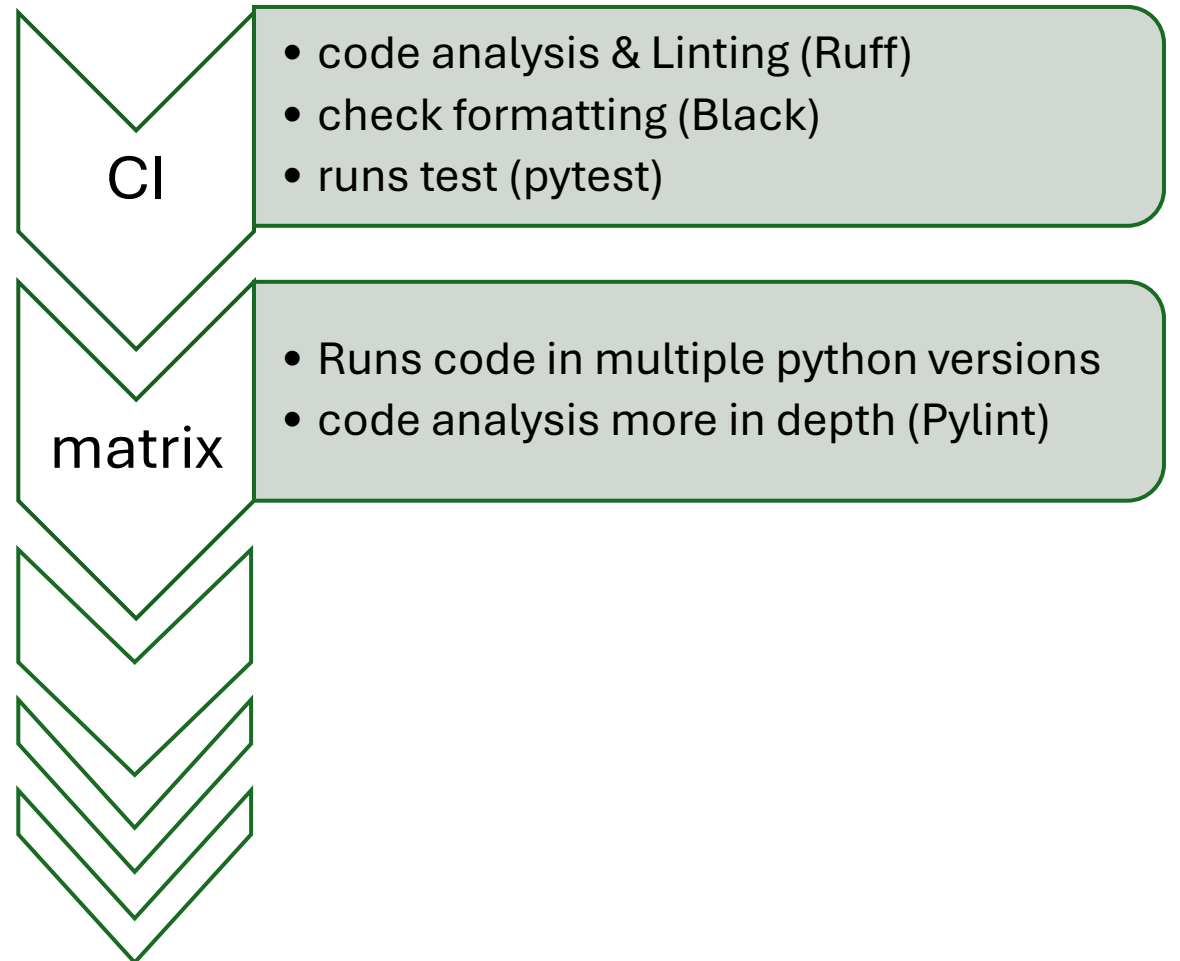


Continuous Integration

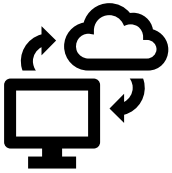
Tracked lines	Covered	Partial	Missed	Coverage %
362	362	0	0	100.00%



- GitHub Actions triggered on:
 - push to main
 - every Pull Request
- If tests pass, Pytest runs:
 - 145 unit tests + doctests
 - Downloadable test artifacts & PR reports
- Coverage threshold:
 - > 80%
 - 1% tolerance from main branch allowed
- Matrix testing (for different versions):
 - Python 3.10 / 3.11 / 3.12
- Pylint tests with minimum score of 8/10



Continuous Deployment



- Automated release workflow via version tags
- Standalone executables built for:
 - Linux
 - Windows
 - macOS
- Automatic GitHub Releases
- Downloadable binaries & checksums
- No Python installation required for .exe
- Dependabot:
 - weekly dependency checks
 - automatic update PRs

build_
binarys

- Creates binaries for multiple OS (PyInstaller)
- Creates GitHub release

▼ Assets 8

-  [butterfly-classifier-linux](#)
-  [butterfly-classifier-linux.sha256](#)
-  [butterfly-classifier-macos](#)
-  [butterfly-classifier-macos.sha256](#)
-  [butterfly-classifier-windows.exe](#)
-  [butterfly-classifier-windows.exe.sha256](#)
-  [Source code \(zip\)](#)
-  [Source code \(tar.gz\)](#)

Final Version

```
C:\Users\Kai\Documents\butterfly-classifier-windows.exe

Results directory: C:\Users\Kai\AppData\Local\Butterfly-Image-Classification\results
Loading machine learning dependencies. This can take a moment...
Checking dataset...
Loading images...
There are total 832 images in this dataset.
Preparing augmented train/validation/test splits...
Min pixel value after normalization: 0.0, Max pixel value after normalization: 1.0
Saved: C:\Users\Kai\AppData\Local\Butterfly-Image-Classification\results\01_random_images_comparison.png
Saved: C:\Users\Kai\AppData\Local\Butterfly-Image-Classification\results\02_class_6_examples.png

=== Training Baseline Model ===
keras\src\layers\convolutional\base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Epoch 1/30
←[1m104/104←[0m ←[32m←[0m ←[0m←[37m←[0m ←[1m43s←[0m 399ms/step - accuracy: 0.5508 - loss: 1.2646 - val_accuracy: 0.4688
- val_loss: 2.0327 - learning_rate: 1.0000e-04
Epoch 2/30
←[1m 46/104←[0m ←[32m←[0m ←[0m←[37m←[0m ←[0m ←[1m22s←[0m 384ms/step - accuracy: 0.7732 - loss: 0.6361
```

```
scripts
├── __pycache__
├── __init__.py
├── config.py
├── data_download.py
├── data_preprocessing.py
├── evaluator.py
├── model.py
├── train.py
└── visualizer.py
```

Butterfly Image Classification

license MIT CI passing codecov unknown release v0.1.4

Potential Improvements



Testing of .exe on
further OS



Implementing
automatic tests
on binaries



Improve
executable for
better user
feedback
(docker, auto-
closing)



Deeper analysis of
flamegraph (code
in other
programming
languages instead
of python)



Thank you.

Questions?